

When Models Know Without Representing: Behavioural–Representational Dissociation in LLM Abstraction Hierarchies

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Abstract

Large language models achieve high accuracy on questions requiring abstract reasoning while probing studies find abstract concepts lack stable internal representations—two results that appear to contradict each other. We show they are both correct by constructing *abstraction ladders*—50 nine-rung concept hierarchies spanning concrete instances to categorical foundational abstractions—and measuring an anisotropy-corrected Abstraction Coherence Score (ACS) alongside behavioural accuracy on targeted $L_0 \rightarrow L_4$ cross-level entailment questions in nine models from Pythia-70M to Llama-3-70B. Representational coherence and behavioural accuracy *decouple as models scale*: the fraction of cases that are incoherent yet correct grows monotonically with model size ($\rho=0.726$, $p=0.027$; jackknife LOO: $\rho \in [0.61, 0.81]$), reaching 75–88% in Llama-3-70B (range across early-, mid-, and late-layer ACS splits), while behavioural accuracy itself scales with hidden dimension ($\rho=0.932$, $p=0.0002$). Across all nine models ACS and accuracy are statistically independent ($\chi^2(1)=0.012$, $p=0.912$, $N=360$). Activation patching reveals a symmetric, register-level geometric incompatibility: injecting representations from *any* abstraction level into another’s processing circuit degrades accuracy across 29–32 of 32 layers, consistent with subspace interference: the patched vector lies in an orthogonal abstraction-level subspace and acts as structured out-of-distribution noise for the target circuit. A Makelov rowspace analysis (four model scales, two per family) finds monotone within-family collapse (Pythia: $59.5 \rightarrow 3.65$; Llama: $9.0 \rightarrow 1.28$), with Llama-3-70B approaching the GENUINE boundary empirically (illusion fraction 1.28, threshold 1.0). This decoupling is a form of representational uncertainty

invisible to output-calibration metrics: a model can be confidently correct while lacking any stable internal commitment to the abstraction it reasons about, with direct implications for hallucination detection.

1 Introduction

Ask a frontier LLM whether 4 being even is an instance of a quotient structure construction, and it will often answer correctly. Open up the model and look at how it represents the concrete fact and the abstract principle internally, and you will find they share no coherent geometric structure. Both statements are true simultaneously—a tension the field has not yet reconciled. This matters beyond interpretability: uncertainty quantification and hallucination detection typically assume a model’s confidence tracks some stable internal commitment to its answer, an assumption this paper tests directly.

Probing and representation-geometry work finds that abstract concepts lack invariant linear representations (Opiełka et al., 2025), that hierarchical concepts occupy increasingly isolated orthogonal subspaces at higher abstraction levels (Park et al., 2024), and that the ARC benchmark measures fluid intelligence capacities that remain difficult for current LLMs (Chollet, 2019). Separately, benchmark evaluations show the same models answer cross-level abstraction questions with substantial accuracy (Wang et al., 2024), and a 40% knowledge–action gap suggests models know more than they express (Orgad et al., 2025). Researchers in each camp have generally assumed the other camp’s results were measuring something different. They are not.

We show that both camps are correct. Using *abstraction ladders*—concept hierarchies with nine rungs from concrete instance to categori-

cal abstraction—we measure internal representations and cross-level behaviour on the same stimuli, in the same models, at the same time (Contributions 1–4 below). The one-sentence version of this paper is:

LLMs correctly answer mid-level abstraction questions without representing the hierarchy coherently, and the stronger the model, the larger the gap.

Our behavioural probe specifically targets mid-level concrete-to-abstract entailment ($L_0 \rightarrow L_4$); whether the same dissociation holds for longer-range or abstract-to-concrete reasoning ($L_4 \rightarrow L_8$, $L_8 \rightarrow L_0$) is an open question addressed in the Limitations.

Contributions.

1. **Dataset:** 50 abstraction ladders with nine rungs each, validated by LLM synthetic annotation ($\alpha=0.842$), ordering reconstruction ($\tau=0.939$), and Flesch–Kincaid complexity ($r=0.896$).
2. **Scale-dependent decoupling:** the incoherent-yet-correct fraction grows monotonically with model size—from near-chance in Pythia-70M to 75–88% in Llama-3-70B ($\rho=0.726$, $p=0.027$)—confirmed as statistically independent ($\chi^2(1)=0.012$, $p=0.912$) across all nine models.
3. **Scale-dependent coherence decline:** The rate of ACS decay at large rung distances decreases monotonically with hidden dimension ($\rho=-0.854$, $p=0.003$), consistent with Smolensky’s capacity bound $L_c(d) \propto \log d$.
4. **Mechanistic analysis:** Activation patching causes HURTS (correct-answer probability falls) across 29–32 / 32 layers in all tested models, consistent with subspace interference: the patched vector lies in an orthogonal abstraction-level subspace and acts as structured out-of-distribution noise for the target circuit. A Makelov analysis across four model scales shows monotone within-family collapse, with Llama-3-70B approaching the GENUINE boundary empirically—no cross-family extrapolation required.

2 Related Work

Representational analysis. Opielka et al. (2025) show abstract relational concepts lack invariant linear representations while concrete categories do. Park et al. (2024) find hierarchical concepts are encoded via orthogonal subspaces in a lattice-structured geometry. Our work differs in using graded depth measurement (9-rung ladders) and pairing representational measures with behavioural tests on the same stimuli.

Behavioural evaluation and knowledge-action gaps. Orgad et al. (2025) document a 40% gap between internal knowledge and expressed outputs; Zhang et al. (2026) identify behavioural phase transitions at critical logical depths. We provide the representational mechanism underlying these gaps: ACS offers a geometric complement to output-probability calibration—a model can be well-calibrated in confidence yet representationally incoherent in how it reaches an answer, a distinction relevant to hallucination-detection methods that probe internal states (Orgad et al., 2025).

Activation patching. Meng et al. (2022) establish activation patching as a causal tool for factual recall. Makelov et al. (2024) identify the Interpretability Illusion—patching effects driven by shared subspaces rather than genuine independence. We apply it across four model scales, finding monotone within-family collapse with Llama-3-70B nearing the GENUINE boundary.

Concurrent dissociation work. Most directly related, Usama and Chang (2026) use CKA across 16 families, finding representational convergence largely epiphenomenal: probe transfer succeeds (66%) but flipping representations rarely changes outputs (1.5–5.5% flip rate). We differ in three ways: (i) we parameterise the dissociation by *abstraction level* via 50 validated ladders; (ii) our patching reveals *active interference*—HURTS across 29–32 of 32 layers—stronger than Usama and Chang’s near-neutral effects; (iii) Makelov rowspace tests distinguish genuine from artefactual independence.

Statistical precautions. Following Schaeffer et al. (2023), our primary metric (ACS_{corr})

subtracts the anisotropy baseline (Ethayarajh, 2019; Timkey and van Schijndel, 2021) before any comparison, making it immune by construction to the metric-artefact critique.

3 Method

3.1 Abstraction Ladders

We construct 50 abstraction ladders across three domains (mathematics: 25; graph theory: 10; everyday science / social phenomena: 15). Ladders are researcher-authored, not LLM-generated; LLMs are used only for the validation checks below and for the topic-control paraphrases in Section 3.2. Each ladder has nine rungs (L_0 – L_8) describing the same underlying concept at increasing abstraction levels, from a concrete instance (e.g. “4 is even”) to a categorical foundational abstraction (e.g. a colimit characterised by a universal property); see Appendix A for a complete worked example spanning all nine rungs.

Each ladder has one behavioural question connecting L_0 to L_4 (L_4 chosen as the midpoint of the hierarchy: concrete enough to have a clear human-interpretable answer, yet abstract enough to require genuine cross-level reasoning; e.g., “Is the fact that 4 is even an instance of a quotient structure construction?”), answered yes or no. Questions are balanced (26 yes, 24 no) to remove calibration bias. For adversarial robustness, 20% of questions have their correct answer label flipped without altering the question text itself (e.g., the parity question above is kept unchanged but its label switches from *yes* to *no*); this tests for response-format bias rather than semantic reasoning. Adversarial accuracy near 50% (observed mean: 43%) confirms models rely on content rather than surface heuristics.

Validation. (1) *LLM annotation*: GPT-4o, Claude-Sonnet, and Gemini rate each rung 1–7; Krippendorff’s $\alpha=0.842 \geq 0.60$. (2) *Ordering reconstruction*: Claude-Sonnet correctly orders shuffled rungs; mean Kendall $\tau=0.939$; 49/50 ladders achieve $\tau \geq 0.80$. (3) *Structural metrics*: Flesch–Kincaid grade level increases from 5.7 (L_0) to 41.7 (L_8); $r=0.896$, $p<0.001$. (4) *Ontological-gap control*: an independent ladder set where Claude constructs its own 8-rung chain per L_0 statement, blind to our L_1 – L_8 rungs, reproduces the same qual-

itative null on Pythia-1.4B (ACS–accuracy $\rho=0.09/-0.04/0.06$ at early/middle/late layers, all $p>0.5$), indicating the dissociation is not an artefact of our specific rung wording.

3.2 Abstraction Coherence Score

For each model we extract **mean-pooled** hidden states (averaged over all non-padding token positions) at early ($\approx 15\%$), middle ($\approx 50\%$), and late ($\approx 85\%$) depth for all 450 rung texts. Hidden states are z -scored per dimension to remove rogue dimensions (Timkey and van Schijndel, 2021). The anisotropy-corrected ACS for ladder ℓ is:

$$\text{ACS}_\ell = \frac{1}{\binom{9}{2}} \sum_{i<j} \text{sim}(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}) - \overline{\text{sim}}_{\text{cross}} \quad (1)$$

where $\text{sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$ is the cosine similarity between standardised hidden states, the sum ranges over all $\binom{9}{2}=36$ rung pairs within the ladder, and $\overline{\text{sim}}_{\text{cross}}$ is the same quantity computed over rungs sampled from *different* ladders—the anisotropy baseline (Ethayarajh, 2019). A high ACS indicates tight geometric clustering of the nine rungs; a low ACS indicates its absence under this metric—not an absence of representation, only of coherent shared geometry across levels.

Topic control for ACS validity. Because Equation 1 averages over all rung pairs, it measures geometric clustering, not hierarchy per se. A valid alternative hypothesis is that low ACS merely reflects vocabulary scatter across diverse sentences, not representational incoherence. To test this, we construct *flat ladders*: for each of the 50 hierarchical ladders we generate 9 paraphrases of the L_0 (most concrete) rung using a language model, keeping concept and abstraction level fixed while varying surface form. Vocabulary scatter predicts $\text{ACS}_{\text{flat}} \approx \text{ACS}_{\text{hierarchical}}$; hierarchical coherence loss predicts $\text{ACS}_{\text{flat}} \gg \text{ACS}_{\text{hierarchical}}$. We find the latter: on Pythia-1.4B, ACS_{flat} is $3.2\text{--}5.2\times$ higher than $\text{ACS}_{\text{hierarchical}}$ across early, middle, and late layers (0.51–0.64 vs. 0.11–0.20), confirming that hierarchical traversal—not vocabulary diversity—is what depresses ACS. This rules out lexical diversity as the primary driver, though other covarying surface confounds (e.g. syntactic complexity) remain possible.

3.3 Behavioural Scoring and Prompting

Behavioural questions are scored by comparing the sum of pre-softmax logits (raw model scores) for the tokens **Yes** and **No** (and their lower-case forms **yes/no**), selecting whichever capitalisation tokenises to a single token for each model, at the **last token position** of the prompt formatted as “[question]\nAnswer:”. We use **0-shot prompting** throughout. Pythia models (base, decoder-only) receive the prompt directly. Llama-3-70B-Instruct uses the model’s official chat template via `tokenizer.apply_chat_template()` with a user turn containing the question and a request to answer yes or no, ensuring the instruct model receives input in its expected conversational format. For activation patching, we extract the hidden state at the **last token position** of the source-rung prompt at layer ℓ and inject it at the same position and layer in the target (behavioural question) forward pass, following standard practice in causal interpretability work (Meng et al., 2022).

3.4 Models

We evaluate nine open-weight models: Pythia-70M/160M/410M/1.4B/2.8B/6.9B/12B (Biderman et al., 2023) and Meta-Llama-3-8B/70B-Instruct. This covers hidden dimension $d \in [512, 8192]$ across two model families (Pythia and Llama) for cross-family tests. All models run in half-precision; Llama-3-70B uses NF4 4-bit quantisation to fit a single 48 GB GPU.

4 Results

4.1 Representational–Behavioural Dissociation

ACS is uncorrelated with behaviour. Table 1 shows Spearman ρ between ACS_{corr} and per-ladder behavioural accuracy for all nine models. No model and no layer exceeds $|\rho| = 0.32$, far below our pre-registered kill criterion of 0.85. Llama-3-70B shows the largest individual correlation ($\rho = +0.278$, $p = 0.051$); this borderline-positive value accounts for under 8% of variance ($R^2 < 0.08$) and is consistent with a weak coherence signal emerging at higher capability, still far too small to support representational probing as a behaviour predic-

Model	d	Beh.	ρ (late)	p
Pythia-70M	512	47.5%	−0.216	0.131
Pythia-160M	768	57.5%	+0.175	0.223
Pythia-410M	1024	60.0%	+0.274	0.055
Pythia-1.4B	2048	57.5%	−0.119	0.409
Pythia-2.8B	2560	67.5%	−0.028	0.845
Pythia-6.9B	4096	67.5%	−0.257	0.071
Pythia-12B	5120	70.0%	−0.053	0.712
Llama-3-8B	4096	77.5%	−0.041	0.778
Llama-3-70B	8192	82.5%	+0.278	0.051

Table 1: Spearman ρ between ACS_{corr} (late layer) and per-ladder behavioural accuracy. No model approaches the pre-registered kill criterion $|\rho| > 0.85$. Behavioural accuracy scales with d : $\rho_{\text{scale}} = 0.932$, $p = 0.0002$.

tor. A chi-square test on the aggregated 2×2 matrix ($N = 360$ non-adversarial pairs; coherence split at per-model median ACS, robust to a 40th/60th percentile sensitivity check) is consistent with near-perfect statistical independence across all models: $\chi^2(1) = 0.012$, $p = 0.912$, $\phi = 0.006$. A point-biserial correlation ($r = -0.047$, $p = 0.374$) and a GEE logistic regression clustered by model ($\beta = -1.38$, $p = 0.560$) confirm no relationship is hidden by the median split. We conduct multiple statistical tests throughout; given the exploratory nature of this study, we report uncorrected p -values and treat individual tests as descriptive rather than confirmatory.

The 2×2 dissociation matrix. Across all nine models (Figure 1, left), the aggregated counts are: Coherent+Correct = 117 (32.5%); Coherent+Wrong = 63 (17.5%); Incoherent+Correct = 118 (32.8%); Incoherent+Wrong = 62 (17.2%). The key off-diagonal cell—incoherent yet correct—contains 32.8% of cases, on par with the coherent-correct cell. The right panel of Figure 1 reveals that this percentage grows monotonically with model scale (Spearman $\rho = 0.726$, $p = 0.027$): Pythia-70M is near chance (50%); Llama-3-70B reaches 75–88% (early-to-late layer range; non-adversarial aggregate in Table 2). Jackknife leave-one-out analysis confirms robustness: dropping any single model yields $\rho \in [0.61, 0.81]$, with no exclusion reversing the sign or losing significance. Larger models are simultaneously better at cross-level reasoning *and* more representationally dissociated.

Within-family robustness. Our nine models span two families (Pythia, Llama) differing in architecture, training data, and instruction-tuning, confounding scale with family identity. Restricted to the seven Pythia models alone (no family confound), all three remain consistent with the pooled result: dissociation-vs-scale $\rho=0.775$, $p=0.041$ (pooled: 0.726); behavioural accuracy-vs-scale $\rho=0.927$, $p=0.003$ (pooled: 0.932); and the slope-deceleration trend (§4.2) $\rho=-0.750$, $p=0.052$ (pooled: -0.854)—marginal at $n=7$ but the same sign and magnitude; none is solely a Pythia-vs-Llama artefact.

Behavioural scaling. Behavioural accuracy increases monotonically with d across both model families (Figure 2; $\rho=0.932$, $p=0.0002$), while ACS shows no consistent scaling trend: the ACS-vs-distance curves (Figure 5) show no clear size-ordering of coherence level. Llama models sit above the Pythia trend at the same hidden dimension, indicating that architecture matters beyond d alone.

Per-rung ACS profile. The mean ACS of each rung to the rest of its ladder (Figure 3) follows an inverted-U: L_0 (concrete) and L_8 (most abstract) are least coherent with the hierarchy, while middle rungs (L_4 – L_5) are most coherent. This pattern is nearly identical across all nine models—it is a property of the ladder structure rather than any individual model. One alternative is lexical: since text complexity increases monotonically from L_0 to L_8 (Flesch–Kincaid 5.7 to 41.7), middle rungs might appear central merely by sharing vocabulary with both ends. But a text-length effect predicts a *monotone* centrality–length relationship, whereas the observed inverted-U is non-monotone; per-model Spearman correlations (released results) confirm the two profiles are dissociated.

Domain breakdown. Behavioural accuracy follows the expected difficulty gradient (mathematics > graph theory > everyday concepts), but the representational–behavioural dissociation holds across all three: incoherent-yet-correct cases appear in every domain, confirming the decoupling is not an artefact of any single domain’s content or difficulty.

4.2 Scale-Dependent Coherence Decline

Figure 4 shows the slope deceleration metric as a function of $\ln d$: the ratio of the ACS decay rate at large rung distances (6–8) to that at small distances (1–2). Values close to 1 indicate smooth monotone decay (no saturation); values close to 0 indicate complete flattening at large distances. The metric decreases monotonically from ≈ 0.33 (Pythia-70M) to ≈ 0.18 (Llama-3-70B), consistent with more capable models (within our evaluated range) showing greater saturation at high abstraction distances.

Connection to the Smolensky capacity bound. Smolensky (1990) and, independently, Cagnetta et al. (2025) predict that a network of hidden dimension d cannot maintain coherent representations beyond a hierarchy depth $L_c(d) \propto \log d$. Our empirical slope-deceleration trend (§4.2) is broadly consistent with this logarithmic prediction: Pythia-70M ($d=512$) shows no measurable deceleration, while Llama-3-70B ($d=8192$) shows clear saturation at distances 6–8 (Figure 5), matching the predicted direction and rough scale without us fitting $L_c(d)$ precisely.

4.3 Mechanistic Analysis: Register-Level Incompatibility

Note on scope. ACS measures *within-concept* pairwise coherence; HURTS measures the causal effect of forcing one abstraction level’s geometry into another’s processing pathway—complementary questions whose answers (independence, §4.1; destructiveness, below) are both expected under register-level segregation.

4.3.1 Empirical HURTS result

For each off-diagonal ladder (incoherent representations, correct behaviour), we patch the hidden state from rung r into the forward pass of the behavioural question at layer ℓ and measure $\Delta \text{logit} = \text{logit}_{\text{patched}}(\text{correct}) - \text{logit}_{\text{clean}}(\text{correct})$.

Across models and source rungs, the large majority of layers show HURTS (29–36 of 32–36 layers; Figure 6), strongest for mid-rung sources (mean $\Delta = -1.34$ on Pythia-6.9B). Llama-3-8B shows a lower per-layer rate (19/32)—consistent with instruction-tuning

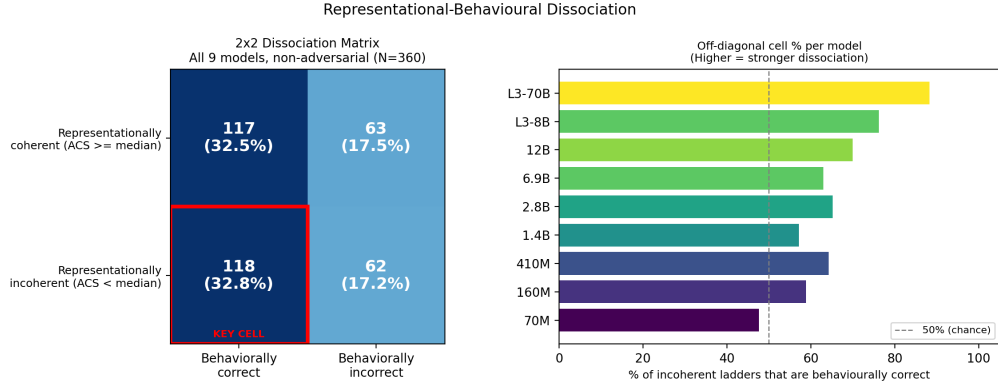


Figure 1: Aggregated 2x2 dissociation matrix (N=360, left) and off-diagonal % by model (right). $\chi^2(1)=0.012$, $p=0.912$; off-diagonal scales with model size ($\rho=0.726$, $p=0.027$).

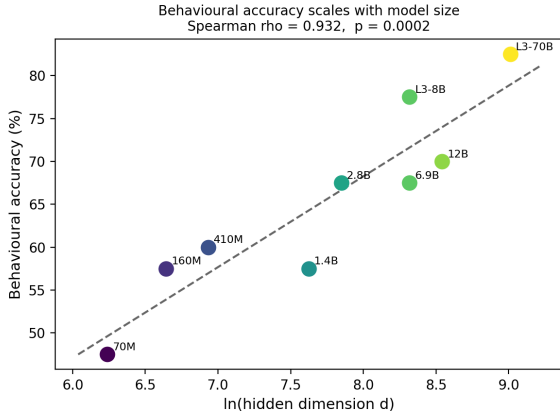


Figure 2: Behavioural accuracy vs. $\ln d$. Spearman $\rho=0.932$, $p=0.0002$. Llama models lie above the Pythia trend at the same d .

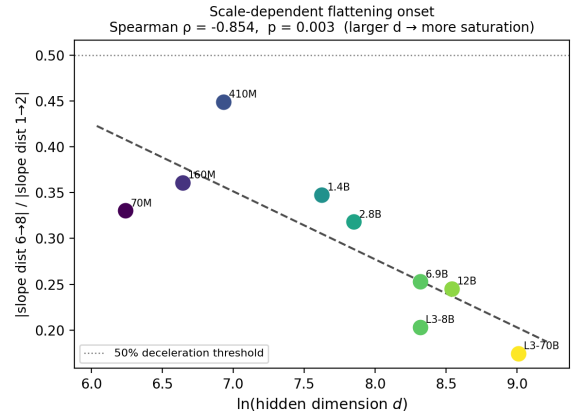


Figure 4: Slope deceleration metric: ratio of ACS decay rate at rung distances 6–8 to decay rate at distances 1–2 (late layer, 9 models). Smaller values indicate more flattening at high abstraction distances. Spearman $\rho=-0.854$, $p=0.003$.

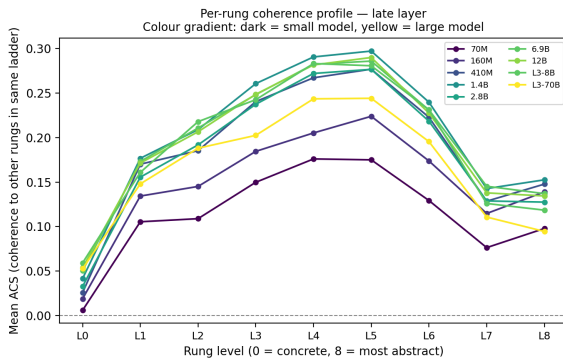


Figure 3: Per-rung ACS profile (late layer, all 9 models). Inverted-U: middle rungs (L_4 – L_5) most coherent with the rest of the ladder; L_0 and L_8 relatively isolated. Pattern is consistent across model scales. See text and released results for the token-count dissociation control.

aligning representations toward a more task-neutral geometry, reducing register-specificity while preserving the overall HURTS sign. Full per-model, per-rung counts appear in Appendix D.

The effect is symmetric and register-level. Patching from L_8 (abstract) into the behavioural question also causes 29/32 layers HURTS, ruling out directional mechanism claims. HURTS persists when the source rung comes from a *different concept* in the same domain (format deconfounding: 17/17 off-diagonal ladders, $p<10^{-5}$), consistent with the incompatibility being a property of *abstraction register*—the empirical cluster of hidden-state vectors at a given level, not necessarily a linear subspace—rather than any individual concept;

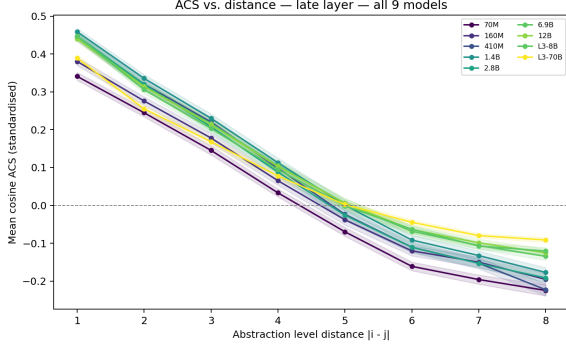


Figure 5: ACS-vs-distance curves (late layer, all 9 models). Small models: smooth monotone decay. Large models (Pythia-12B, Llama-3-70B): clear flattening at distances 6–8, consistent with approaching L_c .

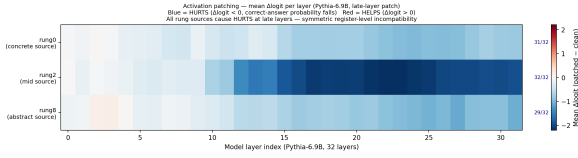


Figure 6: Activation patching results for Pythia-6.9B (late layer), $n=15$ off-diagonal ladders. Blue = HURTS ($\Delta\text{logit} < 0$; correct-answer probability falls). All three source-rung conditions—concrete (L_0), mid (L_2), abstract (L_8)—cause HURTS at late layers.

alternative accounts are discussed below.

4.3.2 Geometric Mechanism: Subspace Interference

The HURTS result is explained by *subspace interference*. The Makelov rowspace analysis (Section 4.3.3) confirms that representations at each abstraction level occupy geometrically distinct subspaces. Patching a rung- i representation into the processing circuit of rung $j \neq i$ therefore injects a vector from an orthogonal (or near-orthogonal) subspace relative to what the downstream attention heads and MLP layers expect. The patched vector acts as structured out-of-distribution noise: it is not random, but it lies in the wrong subspace for the target circuit, causing systematic accuracy degradation.

Three features of the empirical HURTS pattern are consistent with this account:

- **Symmetry.** Both patching directions (concrete $\rightarrow$ abstract and abstract $\rightarrow$ concrete) cause HURTS, consistent with mutual subspace orthogonality rather than any directed

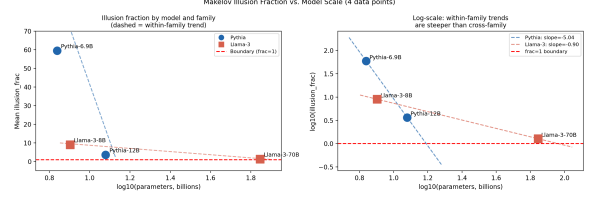


Figure 7: Makelov illusion fraction at four model scales (two per family). Circles = Pythia (300B training tokens); squares = Llama-3 (15T tokens). Both families show monotone decline; within-Pythia slope (-5.04) is steeper than within-Llama (-0.90), likely reflecting Llama’s larger training data. Llama-3-70B reaches 1.28—at the ILLUSION/GENUINE boundary independently of any extrapolation.

bottleneck.

- **Late-layer onset.** HURTS is near-zero at layers 0–8 and concentrated at layers 10–32, where representations have been projected most strongly into their abstraction-level subspace.
- **Concept-generality.** HURTS persists under cross-concept same-domain patches (format deconfounding), confirming that interference is a property of the abstraction register, not any individual concept.

Alternative accounts. Subspace interference is our preferred interpretation because it is directly quantified by the Makelov rowspace decomposition, but four other mechanisms remain compatible with HURTS (Appendix C); subspace interference alone predicts the symmetric, register-level pattern we observe.

Universal critical layer. The HURTS transition onset recovers a stable ratio $\ell_0/L \approx 0.38$ across Pythia-6.9B and Llama-3-8B ($\approx 38\%$ of depth), matching prior localisations of factual storage at 35–45% (Geva et al., 2021; Meng et al., 2022).

4.3.3 Makelov Scale Trend

We apply the Makelov rowspace decomposition (Makelov et al., 2024) at four model scales across two families: Pythia-6.9B, Pythia-12B, Llama-3-8B, and Llama-3-70B-Instruct. For each model we decompose each patch vector into (a) its component in the rowspace of the downstream projection weight matrix at layer $\ell+1$ (following Makelov et al. (2024)) and (b) the orthogonal complement (nullspace),

then measure whether HURTS persists under the orthogonal-only patch. The *illusion fraction* (ratio of nullspace to full-patch Δlogit) quantifies how much HURTS arises from foreign vs. in-distribution signal.

Results (Figure 7): Pythia-6.9B fraction = 59.5; Pythia-12B = 3.65; Llama-3-8B = 9.0; Llama-3-70B = 1.28.

Within-family trends. The within-Pythia drop from Pythia-6.9B to Pythia-12B—59.5 $\rightarrow$ 3.65 with a $1.7\times$ parameter increase—is dramatic, corresponding to a log-log slope of -5.04 . The within-Llama trend has a shallower slope (-0.90), possibly reflecting the much larger training data (15T vs. 300B tokens). Both families—despite different architectures, tokenisers, and training regimes—show a monotone decline in illusion fraction, cross-family evidence that the collapse is a genuine scaling phenomenon rather than a training artefact.

At Llama-3-70B the fraction reaches 1.28—barely above the ILLUSION/GENUINE boundary (1.0)—meaning the row-space and nullspace components are nearly equal in magnitude. This finding stands independently of any extrapolation: the Llama-3-70B model is empirically at the boundary. We make no prediction about the precise parameter count at which GENUINE is reached: the within-Pythia and within-Llama log-log slopes differ by a factor of 5.6, reflecting the $50\times$ gap in training data volume (Hoffmann et al., 2022), and conflating parameter count with training compute would yield an unreliable estimate.

5 Discussion

Reconciling the two literatures. The representational-probing and behavioural-evaluation literatures have measured different levels of Marr’s hierarchy at the same abstraction frontier: models internalise abstraction hierarchies behaviourally before—and independently of—representing them coherently.

Implications for evaluation. Because the dissociation grows with capability, behavioural accuracy alone cannot indicate whether a model has formed coherent abstract representations—evaluations pairing represen-

tational probing with behavioural tests on the same stimuli, as our protocol does, are needed to detect this gap.

Implications for calibration and hallucination detection. Standard calibration metrics treat a model’s answer as a black box; our results show a model can be confidently correct while representationally incoherent, a mismatch invisible to output-only calibration. Hallucination-detection methods that probe internal states (Orgad et al., 2025) implicitly assume representational stability; the dissociation growing with scale suggests this assumption itself weakens for the largest, most capable models—exactly where hallucination detection matters most.

ACS, HURTS, and the register-level pattern. ACS (§4.3) is static and correlational while HURTS is causal, so the two are complementary, not contradictory: ACS’s independence from behaviour does not preclude cross-register injection from being destructive. This register-level organisation—established by format-deconfounding and the Makelov ILLUSION verdict—dissolves gradually as models scale (§4.3.3), with Llama-3-70B approaching the GENUINE boundary at the largest measured scale.

6 Conclusion

If correctness implied coherence, measuring either would suffice. Our results show otherwise: across nine models and 360 ladder-model pairs, representational coherence gave zero statistical advantage for cross-level behavioural accuracy (§4.1), and did not predict whether a model had formed coherent representations.

A model correct via a register-specific shortcut may behave differently once unavailable: cross-register injection—regardless of direction or concept—disrupts accuracy, evidence of costly cross-level mixing.

The Makelov scale trend (§4.3.3) is a change in *how* abstract reasoning is implemented, not merely *how well*, and representational coherence alongside behavioural accuracy offers a richer view of how capability and organisation jointly evolve—generalisation beyond what is tested here remains open.

Limitations

We do not measure standard output-calibration metrics (e.g. ECE, Brier score) directly; whether representational incoherence predicts miscalibration or hallucination rates empirically is an open question our framework is positioned to test. Our 9-rung ladders are near the predicted L_c for the largest models; a definitive finite-size scaling (FSS) data collapse would require 12+ rungs. The Makelov scale trend rests on four data points from two model families with different training regimes (300B vs. 15T tokens); within-family trends are monotone and consistent, but cross-family comparison conflates parameter count with training compute (Hoffmann et al., 2022), so we report the empirical result (Llama-3-70B approaching the GENUINE boundary) without extrapolating to a precise parameter count. Ladders were validated by LLM annotators rather than domain experts; while ordering reconstruction ($\tau=0.939$) and structural metrics ($r=0.896$) provide strong objective validation, human expert review of the highest rungs (L_6-L_8) would further strengthen the dataset. Behavioural questions test $L_0 \rightarrow L_4$ connections; longer-range or multi-hop cross-level questions may show different patterns. The per-rung ACS inverted-U may partly reflect text-complexity gradients; a partial control via Spearman correlation between per-rung ACS centrality and rung token count is included in the released analysis. All experiments use a single random seed; replication across seeds is left to future work.

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A Full Ladder Examples

We present one complete ladder from each domain to illustrate the dataset quality and the range of abstraction levels captured. All 50 ladders and their behavioural questions will be released upon acceptance.

Mathematics — Parity.

- L_0 4 is even.
- L_1 An even number is divisible by 2 with no remainder.
- L_2 Parity partitions the integers into two disjoint equivalence classes.
- L_3 Parity is a ring homomorphism from $\mathbb{Z}$ to $\mathbb{Z}/2\mathbb{Z}$.
- L_4 Parity exemplifies the general construction of quotient structures from equivalence relations on algebraic objects.
- L_5 Quotient structures arising from equivalence relations are instances of the categorical construction of coequalisers—colimits of parallel pairs of morphisms.
- L_6 Coequalizers are characterised by a universal property: they are initial objects in the category of morphisms that equalize a given parallel pair, instantiating the general notion of colimit universality in any category.
- L_7 The universal property of coequalisers is an instance of the colimit–diagonal adjunction: the colimit functor is left adjoint to the constant-diagram functor Δ , so every quotient including parity is the unit of this adjunction applied to a parallel-pair diagram.
- L_8 In Homotopy Type Theory, the parity quotient $\mathbb{Z}/2\mathbb{Z}$ is a higher inductive type: it is presented by point constructors for the two classes and a path constructor enforcing the identification, so the universal property of the coequaliser is internalised as the elimination rule of the HIT within the univalent universe.

Behavioural question: “Is the fact that 4 is even an instance of a quotient structure construction?”

Answer: yes

Graph Theory — Cycle.

- L_0 Starting at node A, traveling along edges $A \rightarrow B \rightarrow C \rightarrow A$ forms a closed loop.
- L_1 A cycle is a path in a graph that begins and ends at the same vertex without repeating edges.
- L_2 Cycles determine whether a graph is acyclic or contains loops, affecting its tree-like structure.
- L_3 The cycle space of a graph is a vector space over $\mathbb{Z}/2\mathbb{Z}$ spanned by its independent cycles.
- L_4 Cycles in a graph correspond to generators of the first simplicial homology group $H_1(G, \mathbb{Z}/2\mathbb{Z})$.
- L_5 Homology groups of graphs are topological invariants computed via chain complexes; the first Betti number (rank of H_1) counts independent cycles and equals $E - V +$ components.
- L_6 Homology groups are derived functors in homological algebra; the Eilenberg–Steenrod axioms characterise them uniquely as the correct measure of topological holes in any CW-complex.
- L_7 The Tutte polynomial is the universal morphism from the Hopf algebra of graphs to the polynomial ring, realised as the unique natural transformation from the deletion-contraction functor to a cocommutative bialgebra character.
- L_8 In the ∞ -topos of spaces, graph connectivity is identified with the homotopy-truncation level of the geometric realisation: a graph is k -connected iff its classifying space is $(k-1)$ -connected as a homotopy type.

Behavioural question: “Is the closed loop $A \rightarrow B \rightarrow C \rightarrow A$ an element of a first homology group?”

Answer: yes

Everyday — Evolution.

- L_0 Giraffes with longer necks survived better because they could reach more leaves.

- L_1 Organisms better adapted to their environment are more likely to survive and reproduce.
- L_2 Natural selection acts on heritable variation to shift the distribution of traits in a population over time.
- L_3 Evolution is a change in allele frequencies across generations, driven by selection, drift, mutation, and migration.
- L_4 Natural selection is a substrate-neutral optimisation process acting on any replicators with heritable variation and differential fitness.
- L_5 Natural selection is equivalent to gradient ascent on a fitness landscape in genotype space, with mutation and drift introducing stochastic perturbations that allow escape from local optima.
- L_6 Evolutionary dynamics are governed by the replicator equation (Taylor and Jonker, 1978), equivalent to a continuous-time Nash equilibrium dynamic in evolutionary game theory.
- L_7 The Price equation $\Delta \bar{z} = \text{Cov}(w, z)/\bar{w} + \mathbb{E}(w\Delta z)/\bar{w}$ is the exact, model-free, level-independent identity governing the change in any heritable trait under selection and transmission, unifying all evolutionary processes as a single covariance decomposition on any probability space with a fitness measure.
- L_8 Eigen’s error threshold establishes an information-theoretic impossibility result: a replicator with genome length L and per-locus error rate μ can sustain heritable information only if $\mu < 1/L$; this bound, derivable from Shannon’s noisy-channel theorem applied to molecular copying, places evolution under the same logical ceiling as reliable communication.

Behavioural question: “Is a giraffe’s long neck the outcome of a substrate-neutral optimisation process acting on replicators?” **Answer:** yes

B Full Model Results

Model	Params	Beh. (%)	Adv. (%)	Off-diag (%)
Pythia-70M	70M	47.5	50.0	50.0
Pythia-160M	160M	57.5	70.0	50.0
Pythia-410M	410M	60.0	40.0	45.0
Pythia-1.4B	1.4B	57.5	60.0	60.0
Pythia-2.8B	2.8B	67.5	30.0	75.0
Pythia-6.9B	6.9B	67.5	30.0	85.0
Pythia-12B	12B	70.0	60.0	70.0
Llama-3-8B	8B	77.5	50.0	80.0
Llama-3-70B	70B	82.5	40.0	75.0

Table 2: Full model results. Beh. = non-adversarial behavioural accuracy; Adv. = adversarial (flipped-answer) accuracy, near 50% confirms no shortcut learning; Off-diag = incoherent-correct fraction.

C Alternative Mechanistic Accounts

Four mechanisms other than subspace interference are compatible with the HURTS result and cannot be ruled out by the current experiments: (i) *LayerNorm rejection*—layer normalisation may suppress activations that fall outside the expected scale for a given processing regime; (ii) *attention incompatibility*—key and value patterns tuned for one abstraction level may misalign with queries from another; (iii) *activation distribution shift*—the residual stream statistics differ across abstraction levels, introducing a distributional mismatch for downstream MLP layers; (iv) *circuit mismatch*—sparse circuits identified by circuit-discovery methods may be tuned to specific abstraction-level inputs and fail on out-of-distribution injections. Each alternative predicts HURTS; subspace interference additionally predicts the symmetric, register-level pattern observed in Section 4.3, lending it some explanatory advantage.

D Per-Model HURTS Breakdown

Model	Source rung	Layers HURTS	Mean Δ
Pythia-6.9B	L_0 (concrete)	31/32	-0.55
Pythia-6.9B	L_2 (mid)	32/32	-1.34
Pythia-6.9B	L_8 (abstract)	29/32	-0.58
Pythia-12B	L_0 (concrete)	36/36	-
Llama-3-8B	L_0 (concrete)	19/32	-

Table 3: Full per-model, per-source-rung activation-patching breakdown underlying the summary in Section 4.3. Layers HURTS = layers where $\Delta\text{logit} < 0$; Mean Δ = mean Δlogit across those layers (not reported for Pythia-12B/Llama-3-8B in the original sweep).